

Solve multi-step addition and subtraction problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A venue has 125,000 seats. It sells 48,675 morning tickets and 39,890 evening tickets. How many remain? Copy or complete the first step.

2. A charity target is £500,000. It raises £187,650 then £204,875. How much more is needed?

3. A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?

4. Miro says: Miro subtracts the two ticket sales separately from the original but loses track of the first subtotal. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A venue has 125,000 seats. It sells 48,675 morning tickets and 39,890 evening tickets. How many remain?
Copy or complete the first step.
Answer 36,435 seats.
2. A charity target is £500,000. It raises £187,650 then £204,875. How much more is needed?
Answer £107,475.
3. A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?
Answer 2,390,850 litres.
4. Miro says: Miro subtracts the two ticket sales separately from the original but loses track of the first subtotal. Is this correct? Explain.
Answer Combine the tickets first or label each new subtotal before continuing.

Solve multi-step addition and subtraction problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A venue has 125,000 seats. It sells 48,675 morning tickets and 39,890 evening tickets. How many remain?

6. Check this result using an inverse, estimate or known fact: A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?

2. A charity target is £500,000. It raises £187,650 then £204,875. How much more is needed?

7. Create a three-step addition and subtraction problem with final answer 250,000.

3. A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?

8. Identify and correct this misconception: Miro subtracts the two ticket sales separately from the original but loses track of the first subtotal.

4. Solve this independently and show every stage: A venue has 125,000 seats. It sells 48,675 morning tickets and 39,890 evening tickets. How many remain?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A charity target is £500,000. It raises £187,650 then £204,875. How much more is needed?

10. Write and solve a similar question to: A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?

Teacher answers and checking prompts

1. A venue has 125,000 seats. It sells 48,675 morning tickets and 39,890 evening tickets. How many remain?
Answer 36,435 seats.
2. A charity target is £500,000. It raises £187,650 then £204,875. How much more is needed?
Answer £107,475.
3. A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?
Answer 2,390,850 litres.
4. Solve this independently and show every stage: A venue has 125,000 seats. It sells 48,675 morning tickets and 39,890 evening tickets. How many remain?
Answer 36,435 seats.
5. Use a second method or representation for: A charity target is £500,000. It raises £187,650 then £204,875. How much more is needed?
Answer Expected result: £107,475. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?
Answer Expected result: 2,390,850 litres. A valid check must be shown.
7. Create a three-step addition and subtraction problem with final answer 250,000.
Answer Answers vary. Every step must be labelled and accurate.
8. Identify and correct this misconception: Miro subtracts the two ticket sales separately from the original but loses track of the first subtotal.
Answer Combine the tickets first or label each new subtotal before continuing.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve multi-step addition and subtraction problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Create a three-step addition and subtraction problem with final answer 250,000.

6. Create a new example that tests the same idea as: A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?

2. Prove the answer to this question, rather than only calculating it: A venue has 125,000 seats. It sells 48,675 morning tickets and 39,890 evening tickets. How many remain?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A charity target is £500,000. It raises £187,650 then £204,875. How much more is needed?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 2,390,850 litres.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro subtracts the two ticket sales separately from the original but loses track of the first subtotal.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Create a three-step addition and subtraction problem with final answer 250,000.
Answer Answers vary. Every step must be labelled and accurate.
2. Prove the answer to this question, rather than only calculating it: A venue has 125,000 seats. It sells 48,675 morning tickets and 39,890 evening tickets. How many remain?
Answer Expected result: 36,435 seats. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A charity target is £500,000. It raises £187,650 then £204,875. How much more is needed?
Answer Expected result: £107,475. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 2,390,850 litres.
Answer One valid reconstruction is: A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro subtracts the two ticket sales separately from the original but loses track of the first subtotal.
Answer Combine the tickets first or label each new subtotal before continuing.
6. Create a new example that tests the same idea as: A reservoir holds 2,500,000 litres. It loses 384,750 litres and gains 275,600 litres. New volume?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.